

Internet Technologies

Become a Full-Stack Engineer

RUNI CS 2026

This year, we expanded the stack into one that includes not only software, but the entire product and its business implications.

The assignment allows and encourages the use of coding agents.



Final Assignment: Building a Web Product with Business Value

Goal of the Assignment

In the final assignment, you will plan, specify, implement, and present a complete software product, end to end.

The goal is not only to write code, but to think like a real product team: identify a business need, plan a technological solution, build it, test it, deploy it to the cloud, and deeply understand every technical decision you made.

The final submission deadline is September 6, 2026.

Stages of Work

1. Product Selection

Choose a product that can be implemented as a Web application and that has real business significance in the real world.

The product should address a clear need, for example:

- Save time
 - Enable the sale of products/services
 - Improve a business process
 - Help users make better decisions
 - Enable an organization to work more efficiently
-

2. Product Specification Document

Write a specification document that explains the product at the business and product levels.

The document should include:

- What problem the product solves
- Who the users of the product are
- Who the customer is
- What the business objectives of the product are

- Which software capabilities need to be built to enable the business objectives
- What the main processes the product allows users to perform are

For example:

- Registration and login
 - Creating a project
 - Uploading information
 - Managing customers
 - Receiving recommendations
 - Making a payment
 - Generating reports
 - Sharing information with other users
-

3. Software Architecture Planning

Plan the structure of the system at the technical level.

You should explain:

- What components the system will have
- Whether you will use a database
- What the main tables or entities in the database will be
- What pages the application will have
- Which Server Actions or API Routes you will need
- How information will flow between the Frontend, the Backend, and the Database
- What users and permissions will exist in the system
- Which external libraries or services you will integrate, and why

4. Detailed Technical Design

Before implementation, write a detailed technical design document.

The document should include:

- Project folder structure
- Structure of the main components
- Database structure
- Main CREATE / READ / UPDATE / DELETE operations
- API description
- Description of the main business logic
- State management in the application
- Error handling
- Input validation
- Planning the main user experience

The goal is to know what you are building before you start writing code.

5. Product Implementation

Implement the product using:

- Next.js
- TypeScript
- Supabase for the Database, and optionally for Authentication
- Vercel for Deployment

The product must be accessible through a URL, and not only run locally on your computer.

6. Test Specification Document

Write a document explaining which tests should be performed to verify that the product works correctly.

The document should include:

- Tests for the main features
- Tests for invalid inputs
- Tests for central business processes
- Permission tests, if there are different user types
- Database tests
- Edge-case tests
- Basic UI tests

The goal is to demonstrate that you know how to define what it means for the product to “work.”

7. Test Implementation

Implement the tests you planned.

You may use tools such as:

- Vitest
- Jest
- React Testing Library
- Playwright
- Documented manual tests, where appropriate

You are not required to test every line of code, but you are required to test the product's main flows.

8. Basic Scalability

Write a document explaining how your product supports basic scalability.

The document should address topics such as:

- What will happen if there are dozens or hundreds of users
 - Which database queries may become heavy
 - Whether database indexes are needed
 - How unnecessary data loading is avoided
 - Whether pagination is used appropriately
 - Whether there is a proper separation between client-side and server-side responsibilities
 - What limitations exist in the current product
 - What you would improve in a future version to support larger scale
-

9. Basic Security

Write a document explaining what you did to keep the product secure at a basic level.

The document should include:

- How Authentication is implemented
- How Authorization is implemented
- Which actions are allowed only for logged-in users
- How you prevent users from accessing another user's information
- How input validation is performed
- How API calls are protected
- How secrets such as API keys are stored
- What security risks still exist and what you would improve in the future

10. Deployment

You must deploy the product using:

- Vercel for the application
- Supabase for the database

The submission must include:

- A link to the live application
 - A link to the GitHub repository
 - Instructions for running the project locally
 - A short explanation of the required environment variables
-

11. Optional Use of Agents

You are allowed, and even encouraged, to use coding agents or AI tools to advance the work, for example:

- Architecture planning
- Creating components
- Writing tests
- Improving the UI
- Finding bugs
- Writing documentation

However, responsibility for the code is yours.

By the end of the process, you must review all of the code, understand it in depth, and be able to explain why every library, component, method, and technical decision was chosen.

A recommended way to do this is to ask a coding agent to create an internal website or document that explains the entire technical structure of the system: the architecture, the main files, the code behind the flows, the database, the tests, and the technical decisions.

12. Preparing for the Product Presentation

At the end of the project, you will present the product for 10–15 minutes.

During the presentation, you should explain:

- What the product is
- What problem it solves
- Who its users are
- Why it has business value
- How the system is built
- What the architecture is
- What the database looks like
- What the main processes in the product are
- Which tests you wrote
- How you thought about scalability
- How you thought about security
- What you would improve if you had more time

Very important: you need to understand the product inside and out.

It is not enough for the application to work — you need to be able to explain how and why it works.

Imagine a short job interview about work you completed.

What You Need to Submit

1. A link to the application on Vercel
2. A link to the GitHub repository
3. Product specification document
4. Detailed technical design document
5. Test specification document
6. Test code
7. Basic scalability document
8. Basic security document
9. Local run instructions
10. A short 10–15 minute presentation

Important Points

The project is not measured only by the number of features, but by the quality of your thinking.

It is better to build a small, clear, useful, secure, and well-structured product than a large, messy, and unstable one.

We expect to see:

- Product thinking
- Organized technical planning
- Clean code
- Proper work with a Database
- Meaningful tests
- Understanding of basic security
- Understanding of basic scalability
- The ability to explain the system in depth

Submission Deadline

Final submission: September 6, 2026.